AI-Assisted Web-Based Cyber Attack Detection System

SecureWatch • Hybrid Rule + AI Cybersecurity Monitoring Dashboard

Supervisor: Md Rafiqul Islam (Rafiq) — Lecturer, Dept. of CSE, BAUET

Team Members: Md. Tamim Ahmed — ID: 0812310205101031 • Sabbir Mobin Setu — ID: 0812310205101053 • Azmain Khan — ID:0812310205101057

Introduction

Web applications face increasing cyber threats that require intelligent and real-time detection. Secure-Watch is an AI-assisted web attack detection system that monitors website requests, combines rule-based analysis with AI to identify suspicious activities, calculates risk severity, and displays real-time security insights through a React dashboard.

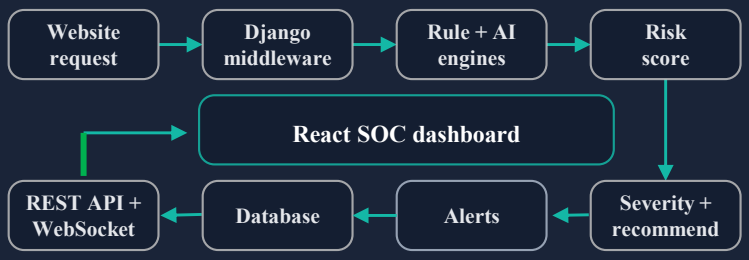
Motivation

Traditional signature systems detect known attacks but can miss changing request patterns. Manual monitoring is difficult because web traffic is large and fast. A hybrid rule + AI approach improves visibility and response.

Objectives

- Capture request data using Django middleware
- Detect SQL Injection, XSS, brute-force and path traversal
- Classify suspicious patterns with an AI model
- Generate confidence, risk score and severity
- Create alerts and mitigation recommendations
- Store logs, threats and alerts in a database
- Visualize live security data in React

Workflow Diagram

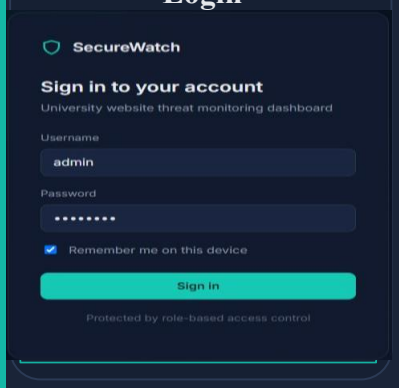


Request metadata is captured, sanitized, analyzed by rules and AI Engine, scored by risk storing engine and displayed instantly through the React SOC dashboard.

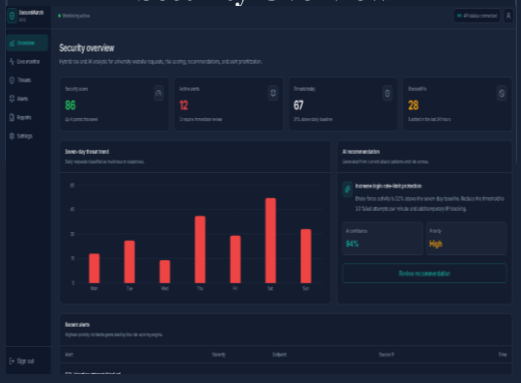
UI/UX Design — React SOC Dashboard

Dark-mode-first interface using navy surfaces, teal actions and severity colors.
Main modules: Login, Overview, Live Monitor, Threats, Alerts, Reports, Settings and Profile.

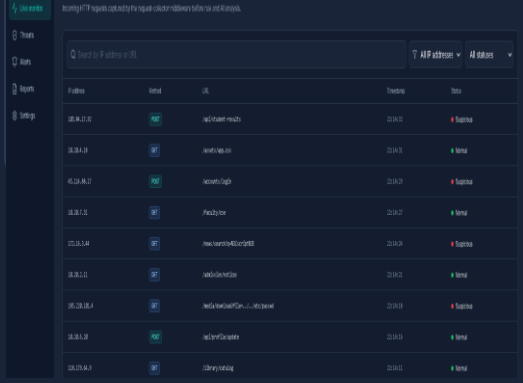
Login



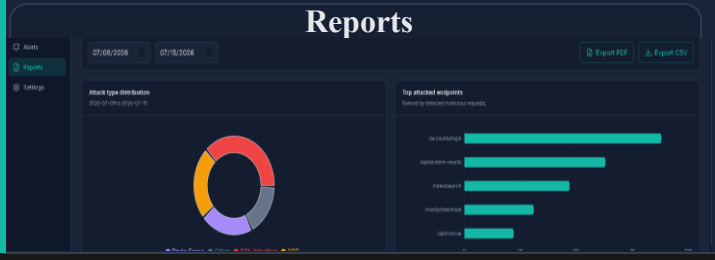
Security Overview



Live Monitor



Reports



Threats & detection



Expected Outcomes

- Functional attack monitoring platform
- Rule + AI threat classification with confidence
- Low / Medium / High / Critical prioritization
- Real-time WebSocket updates and REST APIs
- Persistent records, recommendations and reports

Prototype Status

Works now: detection pipeline, rules, trained demo model, database records, alerts, recommendations, REST APIs and WebSocket updates.

Future deployment: Real university integration, Larger dataset, HTTPS, WAF support, MySQL/Redis and Production hardening.

Technology Stack: React + Tailwind CSS • Django REST Framework • Django Channels/WebSocket • Scikit-learn • Database storage • JWT authentication